

Base-Paper Candidates for a BUET CSE 402 Numerical Analysis / Simulation & Modeling Project

TL;DR

- Six vetted candidate base papers span epidemiology, statistical physics, traffic, sociophysics, finance, and ecology; all map cleanly onto core CSE 401/402 simulation-and-modeling topics (Monte Carlo, Metropolis/MCMC, ODE + Runge-Kutta, discrete-event/cellular automata, curve fitting) and are re-implementable in Python by 5 students in a few weeks.
- The single strongest, safest-yet-impressive choice is the **SIR + Bayesian MCMC epidemic-fitting** track (base paper Grinsztajn et al. 2021 / Chatzilena et al. 2019 on the 1978 English boarding-school flu data), because it combines TWO course pillars (Runge-Kutta ODE solving + Metropolis-Hastings) on a tiny real dataset with abundant open code.
- Runner-ups: the **Nagel-Schreckenberg traffic cellular automaton** (Nagel & Schreckenberg 1992) and the **2D Ising / Metropolis Monte Carlo** model (Metropolis et al. 1953), both classic, cheap to compute, and with rich cross-domain extension paths (traffic engineering, opinion dynamics).

Key Findings

- Every recommended paper is either open-access, on arXiv, or has faithful open reproductions/tutorials, minimizing reproduction risk.
- The simulation-leaning tracks the group prefers (Monte Carlo, MCMC, discrete-event/CA, ODE dynamics) are exactly where the cheapest-to-reproduce yet most "clever" classic papers live.
- Cross-domain framing is easy to bolt on: Ising→opinion dynamics, Gillespie→ecology, Metropolis→image denoising, NaSch→urban traffic policy.
- The biggest reproduction pitfalls are (a) mismatched observation models when fitting real data, (b) statistical convergence/finite-size effects that need careful validation, and (c) papers whose "core" is actually heavy inference machinery — avoided in these picks.

Details

Candidate 1 — SIR/SEIR epidemic model + Bayesian MCMC parameter estimation (MOST RECOMMENDED)

Base paper: Léo Grinsztajn, Elizaveta Semenova, Charles C. Margossian, Julien Riou, "Bayesian workflow for disease transmission modeling in Stan," *Statistics in Medicine* 40(27):6209–6234, 2021. DOI 10.1002/sim.9164; open arXiv:2006.02985. **Secondary/companion:** A. Chatzilena, E. van Leeuwen, O. Ratmann, M. Baguelin, N. Demiris, "Contemporary statistical inference for infectious disease models using Stan," *Epidemics* 29:100367, 2019. DOI 10.1016/j.epidem.2019.100367; arXiv:1903.00423.

Core idea (plain language). Model an epidemic with the classic SIR system of ODEs ($dS/dt = -\beta SI/N$, $dI/dt = \beta SI/N - \gamma I$, $dR/dt = \gamma I$), then treat the transmission rate β and recovery rate γ as unknown parameters to be inferred from real case-count data. Bayesian inference via MCMC produces full posterior distributions (with uncertainty) for β , γ and the basic reproduction number $R_0 = \beta/\gamma$. It is intriguing because it fuses deterministic dynamics with probabilistic inference and yields interpretable epidemiological quantities from a tiny dataset. ([arxiv](#))

Course methods used. ODE solvers (Euler / Runge-Kutta 4 to integrate the SIR system — a direct syllabus item); Metropolis-Hastings / MCMC (core simulation topic); random number generation; least-squares / curve fitting (as a deterministic baseline before going Bayesian); model validation & verification (posterior predictive checks, convergence diagnostics).

Project direction fit. Cross-domain application (epidemiology) AND reproduce+extend — it hits the group's top two preferences simultaneously.

Datasets (small, real, public). (a) 1978 influenza outbreak at an English boarding school: 763 boys at risk, 512 fell ill (a total attack rate of exactly $512/763 = 67\%$), with an epidemic that "lasted from 22 January until 4 February" 1978 initiated by a single returning index case, giving a ~14-point daily "confined to bed" series (1, 3, 6, 25, 73, 222, 294, 258, 237, 191, 125, 69, 27, 11, 4). It was published in the *British Medical Journal*, 4 March 1978 — the digitized original's running header reads "587 BRITISH MEDICAL JOURNAL 4 MARCH 1978," and it is packaged as `influenza_england_1978_school` in the R `outbreaks` package and as `bsflu` in `pomp`. (b) Eyam plague 1665–66: an 8-point S/I table (Raggett 1982, *J. Appl. Stat.* 9:212–225) students can hard-code. (Fabrizio Musacchio ...)

Minimum viable reproduction (Python, 5 students, few weeks). (1) Implement RK4 from scratch to solve SIR; validate against `scipy.integrate.solve_ivp`. (2) Fit β , γ by least squares to the boarding-school data as a deterministic baseline. (3) Hand-code a random-walk Metropolis-Hastings sampler with a Poisson/Negative-Binomial likelihood on daily infected counts; recover posteriors for β , γ , R_0 . (4) Validate: trace plots, burn-in/thinning, acceptance rate tuning, posterior predictive overlay. Expected recovered values: $\beta \approx 1.7/\text{day}$, $\gamma \approx 0.45\text{--}0.5/\text{day}$ (infectious period ≈ 2.2 days), $R_0 \approx 3.5\text{--}4$. **Optional extensions.** Add an Exposed compartment (SEIR) and show R_0 shifts dramatically (SEIR gives $R_0 \approx 10$ on the same data — a great cautionary discussion); apply to the Eyam plague dataset (expect $R_0 \approx 1.5\text{--}1.9$); compare Euler vs RK4 accuracy/runtime. (arxiv) (arxiv)

Difficulty/risk & pitfalls. Low-to-moderate. The base paper uses Stan's HMC/NUTS sampler, not literal Metropolis-Hastings — students substitute a hand-coded MH sampler (the model, likelihood, and data carry over unchanged). Note that a plain SIR/SEIR fits the boarding-school data only imperfectly — this is a documented result, not a failure: as Kucharski/Kalachev et al. put it in "The 1978 English boarding school influenza outbreak: where the classic SEIR model fails" (*J. R. Soc. Interface* 21(220):20240394, 2024), "Previous work has failed to fit classic SEIR epidemic models satisfactorily to the prevalence data of the famous English boarding school 1978 influenza A/H1N1 outbreak." That very tension is excellent analysis material for the report. A minor bibliographic wrinkle: the CRAN `outbreaks` package cites the BMJ source as "1:578" whereas the `epimdr/epidemics` packages and the original scan's header give p.587 — cite p.587.

Citation-graph / helper resources. Official reproduction notebook by Charles Margossian: github.com/charlesm93/disease_transmission_workflow. PyMC official ODE tutorial (Lotka-Volterra, method transfers directly to SIR, demonstrates DEMetropolis): pymc.io/examples/gallery. PyMC3 SIR tutorials (Databricks COVID-19 blog; Towards Data Science). A beginner arXiv note (Okolie, arXiv:2603.15175, 2026) literally derives the SIR likelihood and implements Metropolis-Hastings step by step (non-peer-reviewed preprint, use with care). Related alternatives: Whittles & Didelot (2016), *Proc. R. Soc. B* 283:20160618 (Bayesian Eyam analysis). (arxiv + 3)

Candidate 2 — 2D Ising model via Metropolis Monte Carlo

Base paper: N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, E. Teller, "Equation of State Calculations by Fast Computing Machines," *Journal of Chemical Physics* 21(6):1087-1092, 1953. DOI 10.1063/1.1699114. (One of the most-cited papers in computational science, ~40,000 citations.)

(AIP Publishing)

Core idea. Sample configurations of a lattice of ± 1 spins according to the Boltzmann distribution using the Metropolis acceptance rule (accept a spin flip with probability $\min(1, e^{(-\Delta E/kT)})$). As temperature varies, the system undergoes a phase transition (ferromagnetic $\leftrightarrow$ paramagnetic) at a critical temperature — an emergent collective phenomenon from purely local rules. Intriguing because the same algorithm underlies essentially all of modern MCMC. (Wikipedia)

Course methods. Metropolis-Hastings / Monte Carlo (the canonical example); random number generation; numerical estimation of observables (energy, magnetization, specific heat, susceptibility) via sample averaging; model validation. The validation target is exact: Lars Onsager's 1944 solution ("Crystal Statistics. I.") gives the critical temperature $T_c = 2J / [kB \cdot \ln(1+\sqrt{2})] \approx 2.2692 J/kB$ for the 2D square lattice, with exact critical exponents $\beta = 1/8$, $\gamma = 7/4$, $\nu = 1$.

Direction fit. Reproduce+extend, or cross-domain if framed as opinion dynamics (see below).

Minimum viable reproduction. Implement the 2D Ising Metropolis simulation on an $L \times L$ lattice with periodic boundary conditions; sweep temperature; plot magnetization, energy, specific heat vs T ; locate the critical temperature and compare to the exact Onsager value ≈ 2.269 . Add error bars via block averaging. **Optional extensions.** Implement the Wolff cluster algorithm and compare autocorrelation/equilibration to Metropolis near T_c ; estimate critical exponents ($\beta = 1/8$, $\gamma = 7/4$, $\nu = 1$) via finite-size scaling; **cross-domain:** reinterpret spins as binary opinions (sociophysics) using Stauffer's "Social applications of two-dimensional Ising models" (arXiv:0706.3983) — segregation, opinion consensus, language change. (arxiv)

Difficulty/risk & pitfalls. Low. Very cheap to compute. Pitfalls: critical slowing down near T_c (long autocorrelation with single-spin Metropolis), finite-size effects (need several lattice sizes for scaling), and careful equilibration/burn-in. Abundant tutorials and code (UBC, TU Delft computational-physics courses; many GitHub repos).

Candidate 3 — Nagel-Schreckenberg traffic cellular automaton

Base paper: Kai Nagel, Michael Schreckenberg, "A cellular automaton model for freeway traffic," *Journal de Physique I* 2(12):2221-2229, 1992. DOI 10.1051/jp1:1992277. Open PDF at HAL (hal.science/jpa-00246697).

Core idea. A minimal stochastic cellular automaton: a road is a ring of cells, each empty or holding a car with integer velocity $0 \dots v_{\max}$. Each timestep applies four parallel rules — accelerate, brake to avoid collision, random slowdown with probability p , then move. From these trivially simple rules, realistic traffic jams emerge spontaneously as a collective phenomenon, and the model reproduces the empirical flow-vs-density "fundamental diagram." As the paper's own abstract states: "We introduce a stochastic discrete automaton model to simulate freeway traffic. Monte-Carlo simulations of the model show a transition from laminar traffic flow to start-stop-waves with increasing vehicle density, as is observed in real freeway traffic." Intriguing for how little produces so much.

Course methods. Discrete-event / discrete-time stochastic simulation (core syllabus topic); random number generation; Monte Carlo averaging of observables (flow, mean velocity); model validation against the empirical fundamental diagram; a jamming phase transition around a critical density.

Direction fit. Cross-domain application (physics method → transportation engineering) and comparative benchmarking.

Minimum viable reproduction. Implement the single-lane NaSch model with periodic boundaries; produce space-time diagrams showing jam formation; compute and plot the fundamental diagram (flow vs density) for several $v_{\max}$ and randomization p ; identify the free-flow ↔ jammed transition (unstable fixed point at $p_c \approx 1/(v_{\max}+1)$ in the deterministic limit). **Optional extensions.** Add the Velocity-Dependent-Randomization (VDR / "slow-to-start") rule and show metastability/hysteresis; implement a two-lane model with lane-changing; feed in a real road's density to compare throughput.

[ResearchGate](#)

Difficulty/risk & pitfalls. Low. Extremely light compute. Well documented (Wikipedia, multiple pedagogical PDFs, GitHub gists with Python implementations). Pitfall: careful parallel (synchronous) update ordering; averaging over enough timesteps/realizations to get clean fundamental diagrams; discarding transients.

Candidate 4 — Schelling model of residential segregation (agent-based)

Base paper: Thomas C. Schelling, "Dynamic Models of Segregation," *Journal of Mathematical Sociology* 1(2):143-186, 1971. DOI 10.1080/0022250X.1971.9989794. (Nobel-laureate author; ~4,500+ citations.)

[Taylor & Francis Onl...](#)

Core idea. Agents of two types sit on a grid; each is "unhappy" if the fraction of like neighbors falls below a tolerance threshold, and unhappy agents relocate. Even with only mild same-type preference (e.g., wanting $\geq 30\%$ similar neighbors), the system self-organizes into stark global segregation — a famous "micromotives → macrobehavior" surprise. [arxiv](#)

Course methods. Discrete-event/agent-based stochastic simulation; random number generation; Monte Carlo measurement of segregation indices; model validation (tipping-point / phase-transition behavior). Can be connected to the Ising model (statistical-mechanics analogue).

Direction fit. Cross-domain application (economics/sociology) and reproduce+extend.

Minimum viable reproduction. Implement the grid, tolerance threshold, and relocation dynamics; measure a segregation metric (e.g., average share of like neighbors) vs tolerance threshold; reproduce the emergence of segregation and the tipping point. **Optional extensions.** Multi-dimensional agents; variable/asymmetric group tolerances; connect quantitatively to the Ising/statistical-physics framework (Stauffer & Solomon 2007, "Ising, Schelling and self-organising segregation"; Vinkovic & Kirman 2006 "physical analogue," *PNAS* 103:19261).

Difficulty/risk & pitfalls. Very low difficulty (a classic CS teaching assignment — Stanford Nifty Assignments has a version). Risk that it looks "too easy"/not numerically rich — mitigate by adding quantitative phase-transition analysis and the Ising connection. Best combined with rigorous measurement to satisfy a numerical-methods rubric.

Candidate 5 — Monte Carlo option pricing (finance)

Base paper: Phelim P. Boyle, "Options: A Monte Carlo approach," *Journal of Financial Economics* 4(3):323–338, 1977. DOI 10.1016/0304-405X(77)90005-8. (~1,350+ citations; the paper that introduced Monte Carlo to option pricing.) ([REPEC](#))

Core idea. Simulate many random price paths of an underlying asset under geometric Brownian motion, compute the option payoff on each path, average, and discount to get the option price. For a European call this can be checked against the closed-form Black–Scholes formula; Monte Carlo shines where no closed form exists (Asian, barrier, path-dependent options). Boyle also introduced variance-reduction (control/antithetic variates). ([ScienceDirect](#)) ([ResearchGate](#))

Course methods. Monte Carlo simulation; random number generation (Gaussian sampling); numerical convergence analysis; variance-reduction as a form of numerical efficiency improvement; validation against the Black–Scholes analytic benchmark.

Direction fit. Cross-domain application (quantitative finance) with a clean comparative-benchmarking angle (Monte Carlo vs closed-form; convergence and variance reduction).

Minimum viable reproduction. Simulate GBM paths; price a European call/put by Monte Carlo; verify convergence to Black–Scholes as path count grows; implement antithetic and control variates and quantify variance reduction. **Optional extensions.** Price an Asian or barrier option (no closed form); quasi-Monte Carlo (low-discrepancy sequences); the Longstaff–Schwartz least-squares Monte Carlo for American options (adds regression/curve-fitting — another syllabus topic). ([arxiv](#))

Difficulty/risk & pitfalls. Low-to-moderate. Data-free (parameters are chosen), so no data-availability risk. Pitfalls: correct discounting and risk-neutral drift; enough paths for tight confidence intervals; care with random-seed reproducibility. Widely reproduced with Python tutorials.

Candidate 6 — Gillespie stochastic simulation algorithm (chemical kinetics → ecology/epidemiology)

Base paper: Daniel T. Gillespie, "Exact Stochastic Simulation of Coupled Chemical Reactions," *Journal of Physical Chemistry* 81(25):2340–2361, 1977. DOI 10.1021/j100540a008. (Foundational; tens of thousands of citations.) ([Wolfram Cloud](#))

Core idea. Instead of deterministic ODE rate equations, simulate every discrete reaction event exactly: sample the waiting time to the next event from an exponential distribution and sample which reaction fires proportional to its propensity. This captures intrinsic stochastic fluctuations that matter when molecule/individual counts are small. It is a discrete-event Monte Carlo method with exact statistical correctness. ([Wikipedia](#)) ([Medium](#))

Course methods. Discrete-event simulation (textbook match); Monte Carlo; random number generation (exponential and categorical sampling via inverse-CDF); comparison to the deterministic ODE (Euler/RK) limit — bridging both halves of the syllabus.

Direction fit. Cross-domain application (the algorithm is chemical but transfers to ecology, epidemiology, gene expression) and comparative benchmarking (stochastic vs deterministic). ([eLife](#))

Minimum viable reproduction. Implement the Gillespie direct method for a small reaction network; apply to a stochastic SIR epidemic OR a stochastic Lotka–Volterra predator–prey system; compare stochastic trajectories and their variability to the deterministic ODE solution (which the team also solves

with RK4). Show extinction events / fluctuations invisible to the ODE. **Optional extensions.** Implement tau-leaping for speed and compare accuracy/runtime to the exact method; the Gibson-Bruck next-reaction method. ([PubMed](#))

Difficulty/risk & pitfalls. Low-to-moderate. Data-free. Pitfalls: correct propensity updates, efficiency for larger systems (motivates tau-leaping), and enough realizations to characterize the distribution. Many Python tutorials exist.

Recommendations

Ranked top 3 for this constraint set (simulation-leaning, cross-domain preferred, Python, limited time, 5 students):

1. **SIR + Bayesian MCMC (Candidate 1).** The best fit: it is genuinely cross-domain (epidemiology), and it simultaneously exercises TWO major syllabus pillars — Runge-Kutta ODE integration AND Metropolis-Hastings/MCMC — plus curve fitting and model validation. The dataset is tiny, real, and trivially available; there is a wealth of open reproduction code; and the work naturally partitions across 5 people (ODE solver, MH sampler, data/likelihood, validation/diagnostics, extensions). Safe because a deterministic least-squares fit already gives a working baseline; impressive because the full Bayesian posterior with R0 uncertainty and the SIR-vs-SEIR contrast is publication-flavored. **Start here.**
2. **Nagel-Schreckenberg traffic CA (Candidate 3).** The strongest pure simulation-and-modeling pick: a beloved, minimal, stochastic discrete-time model with a striking emergent result (spontaneous jams) and a clean quantitative validation target (the fundamental diagram + jamming transition). Essentially zero data-availability or compute risk, strong cross-domain (physics → transportation) story, and obvious extensions (VDR, two-lane). Safe-yet-impressive because the core is a weekend implementation, leaving weeks for the interesting phase-transition and policy analysis.
3. **2D Ising / Metropolis Monte Carlo (Candidate 2).** The canonical Monte Carlo/Metropolis paper, with an exact analytic benchmark (Onsager $T_c \approx 2.269$) for validation and a natural cross-domain pivot to opinion dynamics/segregation (sociophysics). Slightly more physics background required, but conceptually central to the course. Choose this if the group wants the "textbook" Metropolis experience with a clever social-science twist.

Decision thresholds. Pick Candidate 1 if the group wants to showcase MCMC + ODE together on real data (highest ceiling). Pick Candidate 3 if they want the lowest-risk, most self-contained simulation with guaranteed clean results. Pick Candidate 2 if they want the definitive Monte Carlo/Metropolis demonstration and enjoy statistical physics. Candidates 5 (finance MC) and 6 (Gillespie) are excellent alternates that also cleanly combine deterministic and stochastic methods; Candidate 4 (Schelling) is the easiest but needs added numerical rigor to satisfy the rubric.

Caveats

- Several "base papers" here are foundational classics (Metropolis 1953, Boyle 1977, Gillespie 1977, Nagel-Schreckenberg 1992, Schelling 1971) rather than recent papers; this is deliberate — they are maximally reproducible and well-documented. If the professor requires a *recent* paper, use a modern application paper that builds on these (e.g., Grinsztajn 2021 for SIR-MCMC, or a recent Ising-sociophysics review such as Mullick & Sen 2025, arXiv:2506.23837).

- Reported parameter values ($R_0 \approx 3.5-4$ for boarding-school flu, $T_c \approx 2.269$ for 2D Ising, critical traffic density $\rho_c \approx 1/(v_{\max}+1)$) are the standard literature values the team should aim to recover; exact numbers depend on the observation model and estimator chosen.
- The base paper for Candidate 1 uses Stan's HMC/NUTS rather than literal Metropolis-Hastings; the team must implement MH themselves to satisfy the syllabus — straightforward but worth flagging up front.
- Citation counts are approximate and drawn from Google Scholar / Semantic Scholar aggregations as of the research date; treat them as orders of magnitude, not exact.
- Always confirm the professor accepts foundational papers and cross-domain reinterpretations before committing.